

* Memory mgmt

It is the process of allocating, using & freeing memory automatically.

create object



memory allocated



program uses object



obj. no longer needed



garbage collector frees

memory

1. Memory Allocation

When we create a variable Python allocates memory automatically.

x = 10

↳ Python store obj in memory.

2. Reference Count

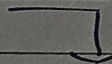
Python keep track how many variable are referencing to obj.

a = [1, 2, 3]

b = a

memory

a



[1, 2, 3]



b

ref. c = 2

3. Garbage Collection

Python automatically remove obj. that are no longer used

x = [1, 2, 3]

sys module ↙

x = None

list has no references

garbage collector remove it from memory

4. memory heap
All Python obj are stored
in heap memory.

int, list, dict, string, tuple.

5. Stack memory

stores:

- Function calls
- local var.
- Execution info.

6. Private heap

- only Python memory
manager can access it.

7. memory manager

Python has built in memory
manager.

- allocate memory
- Tackle obj.